computing
conference 2026

A Multi-Objective Adaptive Learning Algorithm

A Synthesis of Pedagogical Models: From Recommender Systems to Pedagogically Sound Instructional AI

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Learning Style | Difficulty/ZPD | Cognitive Load | Knowledge Gap | Engagement

Why Do Current Adaptive Learning Systems Fall Short?

What Systems Do Now		What Learners Actually Need
✗ **BKT / IRT** Knowledge mastery only		✓ **Knowledge Gap Targeting** Address real weaknesses (BKT)
✗ **Collaborative Filtering** Predicts ratings, not learning	vs	✓ **Cognitive Load Optimization** Prevent overload, enable flow
✗ **Content-Based** Static profiles, no adaptation		✓ **Engagement + Style Match** Motivate and personalize

"Current systems leave 80% of the learning experience unoptimized."

What Does the Literature Tell Us?

Content-Based Filtering	Collaborative Filtering	Proposed ALS ★
✓ High personalization	✓ Learns from peers	✓ Multi-dimensional input
✓ Simple, interpretable	✓ Scalable with data	✓ Dynamic learner model
✗ Static profiles	✗ Cold-start problem	✓ Cognitive load + ZPD
✗ No cognitive state	✗ Opaque recommendations	✓ Prerequisite enforcement
✗ No pedagogical grounding	✗ Ignores learning sequence	✓ Explainable weights
✗ Cold-start safe, but rigid	✗ No ZPD or cognitive load	✓ Real-time BKT updates

★ Our Work Occupies the Gap — Formally Integrated, Constrained, Explainable Adaptive Learning

The Proposed Multi-Objective Adaptive Algorithm

C* = argmax S(c, st, t)

Optimal next content = argmax of weighted scoring function

LS	D	CL	KG	Eng
Learning Style	**Difficulty/ZPD**	**Cognitive Load**	**Knowledge Gap**	**Engagement**
Modality alignment	Zone of Proximal Development	Prevent overload	BKT-based weakness	Sigmoid prediction

+ E Exploration Bonus (UCB)
β·√(ln(Nt)/Nc)

Subject to: Prerequisites(c) ⊆ Mastered(st) | D(c) ∈ ZPD(st) | CL_projected < CL_max

Scoring Function S(c, st, t) — The Mathematical Engine

$$S(c, st, t) = w_{ls} \cdot LS(c, st) + w_d \cdot D(c, st) + w_{cl} \cdot CL(c, st) + w_{kg} \cdot KG(c, st) + w_e \cdot Eng(c, st) + \beta \cdot \sqrt{\frac{\ln(N_t)}{N_c}}$$

LS — **Learning Style**

Weighted affinity between content format & learner modality

D — **Difficulty/ZPD**

Gaussian centered on K(st)+δ — optimal challenge zone

CL — **Cognitive Load**

Minimize deviation from optimal projected load

KG — **Knowledge Gap**

Prioritize unmastered topics via BKT probability

Eng — **Engagement**

Sigmoid prediction of continued engagement

E — **Exploration**

UCB bonus: explore less-used content, exploit best

Component Formulas — Pedagogical Grounding in Math

LS

Learning Style Match

$LS(c,st) = \Sigma\ M(type(c),\ style\_j) \cdot P(style\_j|st)$

Weighted affinity between content format and learner preference distribution

D

Difficulty / ZPD

$D(c,st) = exp(-(d\_c-(K(st)+\delta))^2\ /\ 2\sigma^2\_zpd)$

Gaussian peak at ZPD center — ideal challenge δ above current knowledge

CL

Cognitive Load

$CL(c,st) = 1 - |L\_projected-L\_optimal|\ /\ L\_optimal$

Score=1 when projected load equals optimal; penalizes both over- and under-load

KG

Knowledge Gap

$KG(c,st) = 1 - P(mastery\_topic(c))$

High score for unmastered topics via BKT; 0.5 for new/unseen topics

Eng

Engagement Prediction

$Eng(c,st) = \sigma(\Sigma\ \theta\_i\ \cdot\ \varphi\_i(c,st))$

Sigmoid of learned features: style match, time-of-day, recent performance

E

Exploration Bonus (UCB)

$E = \beta(t) \cdot \sqrt{(ln(N\_total)/(N\_c+1))}$

Reinforcement learning UCB: explore under-used content, β decreases over time

Algorithm Flow — Closed-Loop Adaptive Process

1. Input	2. BKT Update	3. Filter	4. Score	5. Constrain	6. Select C*	7. Deliver	8. Record
Content pool + Student state st	Estimate K(st) from observations	Check prerequisites → C_eligible	Calculate S(c,st,t) for each eligible	Apply ZPD + CL_max bounds	argmax S(c,st,t)	Present optimal content to learner	Log response → repeat loop

↻ **ITERATIVE ADAPTATION — After each response, BKT updates K(st) and the loop restarts in real-time**

Real-time

Per-interaction

Prerequisites

Gating ensures foundational safety before advancing

Multi-Constraint

ZPD + CL_max enforce pedagogical boundaries on every selection

Closed-Loop

Every learner action refines the model — genuinely adaptive

Three Novel Contributions

What makes this approach fundamentally different

01

Formally Constrained Multi-Objective Optimization

- Hard pedagogical constraints: Prerequisites, ZPD, CL_max
- Ensures educational integrity & safety
- Absent in pure RL and CF models

02

Unified Cognitive & Affective State Balancing

- Simultaneous weighting of 5 learning dimensions
- Dynamic, explainable trade-offs
- Tunable to learner phase & curriculum goals

03

Enhanced Explainability & Tunability

- Explicit weights (w_i) educators can read & adjust
- Trace every content decision to its factors
- Unlike opaque Matrix Factorization

Pedagogical Constraints & Future Work

Safety Guardrails — Non-Negotiable Constraints

1. **Prerequisite Mastery**
   $Prerequisites(c) \subseteq Mastered(st)$
   No content served before foundational knowledge is secured

2. **ZPD Envelope**
   $D(c) \in ZPD(st)$
   Optimal challenge maintained — never overwhelming

3. **Cognitive Load Cap**
   $CL\_projected(c,st) < CL\_max$
   Prevents burnout in high-stakes health professions learning

Future Work Roadmap

- **Q2 2026** Physiotherapy pilot implementation
- **Q4 2026** Clinical reasoning outcome measurement
- **Q1 2027** Retention & engagement analysis
- **Q2 2027** Scale to other health professions

Conclusion

BEFORE

- Single-objective optimization
- Black-box recommendations
- Predicts preference or knowledge
- Static, opaque models

→

AFTER

- Multi-objective constrained optimization
- Explainable, tunable decisions
- Optimizes complete learning experience
- Dynamic, pedagogically grounded

A conceptual shift in educational AI — from predicting what students know to optimizing professional competence.

01 Formally constrained multi-objective optimization

02 Unified cognitive & affective balancing

03 Explainable weights — educators can trace & tune

📂 **Explore the Code**
github.com/kwid-ai/multi-obj-adaptive-learning-algorithm

🤝 **Collaborate With Us**
Physiotherapy education pilot — seeking research partners

Thank You

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Questions Welcome!

"What dimension would YOU add to the scoring function?"